

When Does Hierarchical k -Means Curation Beat Random Sampling? The Decisive Role of the Embedding Space

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Abstract

Hierarchical k -means (HKM) curation builds large, diverse and *balanced* datasets for self-supervised pre-training by clustering embeddings and resampling clusters uniformly (Vo et al., 2024). A natural expectation is that, on an imbalanced pool, this selects a better class-balanced subset than uniform random sampling. We test this in a controlled small-scale setting (Imagenette, 10 classes), isolating two factors: the balance of the source pool and the embedding space used for clustering. First, on the naturally balanced pool HKM offers no downstream advantage over random: at a 20% budget it matches or trails random under both linear-probe and fine-tuning evaluation. Second, on a synthetic long-tailed pool whether HKM improves subset balance is decided by the clustering features – supervised ResNet-ImageNet and Places365 embeddings yield more balanced subsets than random, whereas task-agnostic DINOv2 embeddings yield a *less* balanced one. Third, even when HKM improves measured balance this does not reliably transfer to accuracy: random stays competitive or better on overall and worst-class accuracy, with HKM helping tail classes only at the smallest budgets.

1. Introduction

Training-data selection is a lever for both efficiency and model quality: not all examples are equally useful, and pruning or reweighting a pool can match full-data performance at a fraction of the cost (Sorscher et al., 2022). For self-supervised pre-training, Vo et al. (2024) argue that good pre-training data should be *large, diverse and balanced*, and propose hierarchical k -means (HKM): embeddings are clustered with several successive levels of k -means, and a hierarchical sampling step then draws examples so that the selected subset is spread evenly across clusters rather than across raw examples. Because dense regions of the embedding space are downweighted, HKM is expected to counteract the imbalance of an uncured pool. It builds on a line of clustering-based representation work (Caron et al., 2018) and is typically run on task-agnostic features such as DINOv2 (Oquab et al., 2023).

Research question. Does HKM select a better class-balanced subset than random sampling, and under what conditions does any advantage appear? We vary the source pool’s balance and the clustering space independently, and measure subset balance without training before asking whether it transfers to a trained model. Our contributions are:

- **On a class-balanced pool, HKM gives no downstream advantage over random.** Across two evaluators (ResNet-ImageNet, ResNet-Places365) and both regimes at a 20% budget, HKM with DINOv2 features matches or trails random and distorts the class distribution along the embedding geometry (Table 1, Figure 1).
- **The clustering embedding space decides whether HKM out-balances random on a long-tailed pool.** Supervised ResNet-ImageNet and Places365 features

yield subsets more balanced than random; task-agnostic DINOv2 features yield one *less* balanced than random (Figure 2a).

- **Improved subset balance does not reliably improve accuracy.** Despite higher subset entropy than random, HKM-Places365 does not surpass random on overall or worst-class accuracy and helps tail classes only at the smallest budgets (Figure 2b).

2. Method and Experimental Setup

Data. We use Imagenette (Howard, 2019), a 10-class subset of ImageNet (Deng et al., 2009), with $\approx 9,500$ training and $\approx 3,900$ validation images (≈ 950 per class). The pool is naturally near-uniform (normalized class entropy 0.999), a clean baseline for studying curation under controlled imbalance.

Curation by HKM. We use the reference implementation of Vo et al. (2024) (`ssl-data-curation`): given L_2 -normalized embeddings, we build a 3-level hierarchical k -means with resampling and draw a target-size subset by hierarchical sampling (random within-cluster strategy). For the balanced pool `n_clusters` = [1000, 300, 100] and `sample_sizes` = [5, 3, 2]; for smaller long-tailed pools we scale the cluster counts with pool size ($n_1 = \max(N/10, 30)$, then $/3$ per level). Selection is *unsupervised* – HKM sees only embeddings.

Embeddings. We compare three clustering spaces: DINOv2 ViT-L/14 (Oquab et al., 2023), a task-agnostic encoder never trained on Imagenette; ResNet-50 pre-trained on Imagenette (He et al., 2016); and ResNet-50 pre-trained on Places365 (Zhou et al., 2017).

Synthetic long-tail. To create controlled imbalance we subsample the pool to a power law: each class gets a fixed random rank r and is kept to $\min(n_{\max} r^{-\alpha}, n_c)$ images. We use $\alpha \in \{0.5, 1.0, 2.0\}$ (headline $\alpha=1.0$), giving pools of $N = 4932, 2856, 1487$ images whose normalized class entropy drops from 0.972 through 0.873 to 0.549 as the tail steepens.

Balance metrics (no training). For a selected subset we report normalized class entropy $H/\ln K$ (1= perfectly balanced) and KL divergence to the uniform distribution (0= perfect). These measure the subset directly, decoupling selection quality from training variance.

Evaluators. Subset quality is measured by training a ResNet-50 on the selected images and evaluating on the *balanced* validation set, under two initialisations (ImageNet, Places365) and two regimes: a *linear probe* (frozen backbone, only the final layer trained) and *fine-tuning* (all weights). Training uses AdamW (lr 10^{-3} , weight decay 10^{-4}), batch size 64/GPU, 10 epochs. We report overall, worst-class, and tail-class accuracy (mean over the three least-frequent classes). Results use a single seed, so we treat sub-point differences as noise (Section 4).

3. Results

On a balanced pool, HKM does not beat random (negative control). Table 1 compares HKM (DINOv2 curator) and random selection at a $\approx 2,000$ -image budget ($\approx 20\%$), with the full-data model as a ceiling. In every evaluator/regime cell HKM equals or trails random: the gap is small under linear probing (≤ 1 point) but clearly negative under

fine-tuning (-1.8 points for ResNet-ImageNet). Figure 1 explains why. Whereas random leaves every class near the balanced count (189 images at 20%), HKM over- and under-selects classes by the geometry of the curation space, most starkly in DINOv2 space where one class is selected far above the rest. With no imbalance to correct, sampling evenly across clusters mildly *distorts* the class distribution rather than balancing it, motivating the introduction of imbalance.

 Table 1: **Balanced pool (negative control).**

Selection	ResNet-ImageNet		ResNet-Places365	
	Linear probe	Fine-tuning	Linear probe	Fine-tuning
Full pool (ceiling)	0.988	0.989	0.930	0.955
Random ($\approx 2k$)	0.981	0.968	0.911	0.941
HKM-DINO ($\approx 2k$)	0.979	0.950	0.901	0.933

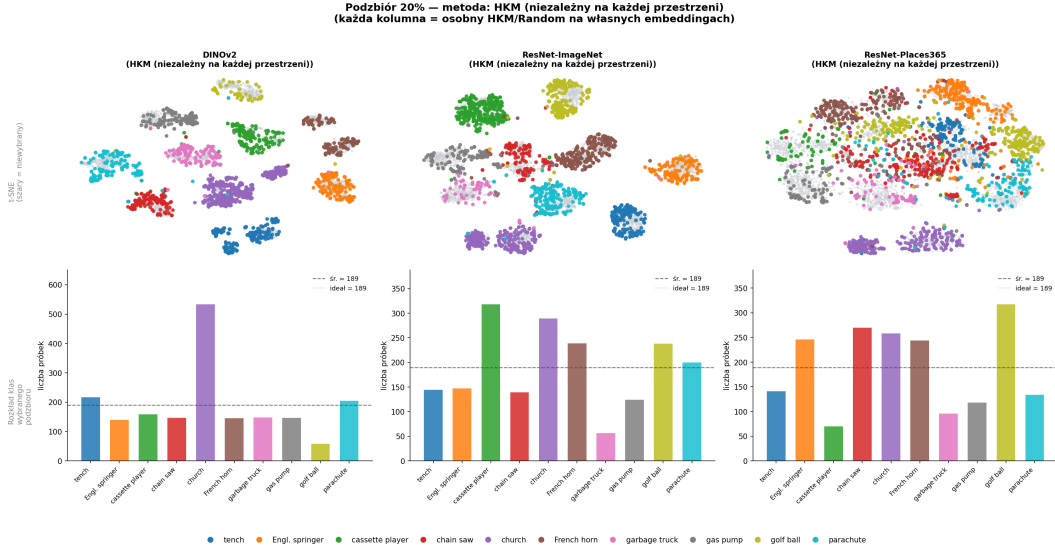


Figure 1: **On a balanced pool HKM is not balance-seeking; it mirrors the curation geometry.**

On a long-tailed pool, the curator decides subset balance. Figure 2(a) measures the balance of the subset *selected* from the headline pool ($\alpha=1.0$) vs. budget, for HKM in three spaces and random. The ordering is consistent and is the central result: HKM-ResNet-ImageNet is the most balanced, HKM-Places365 next, both above random; task-agnostic DINOv2 falls *below* random (normalized entropy ≈ 0.63 vs. ≈ 0.87 at 20%). The balancing effect is thus not intrinsic to the algorithm: in DINOv2 space, evenly covering clusters concentrates rather than balances classes, because tail classes do not form separate dense clusters there.

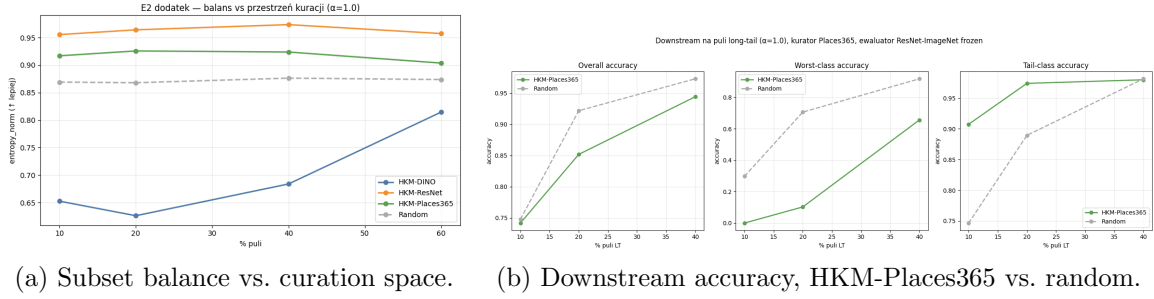


Figure 2: **Long-tailed pool ($\alpha=1.0$): the curator sets subset balance, but better balance does not guarantee a better model.**

Better subset balance does not guarantee a better model. We then take the curator that does improve balance (Places365) and ask whether the more balanced subset yields a better model. Figure 2(b) shows it largely does not: random is competitive or better on overall and worst-class accuracy, and HKM-Places365 beats random only on tail-class accuracy and only at the smallest budgets (10–20%), the gap closing by 40%. A more balanced *count* of tail examples thus does not yield uniformly better *features* for those classes; the extra tail coverage helps where data is scarcest and washes out once the budget lets random cover the tail.

4. Conclusions

Across the three experiments a single picture emerges: the balancing behaviour commonly attributed to HKM is a property of the embedding space, not of the sampling algorithm. On a balanced pool there is nothing to balance, and HKM matches or slightly trails random while distorting the class distribution along the curation geometry (Table 1, Figure 1); on an imbalanced pool HKM out-balances random only in a supervised feature space where classes form separable dense clusters, and under-balances it in the task-agnostic DINOv2 space the original method favours (Figure 2a). Even a genuine balance gain (Places365) does not buy a uniformly better model (Figure 2b). This is consistent with HKM being designed and validated at web scale (Vo et al., 2024), where diversity, not per-class balance on a 10-class proxy, is the goal. The take-away is to choose the curation embedding for the property one cares about: for class balance on a downstream taxonomy, a feature space aligned with that taxonomy out-balances a task-agnostic one – which can do worse than random.

Limitations. The study is deliberately small: a single 10-class dataset, so conclusions need not transfer to the large open-vocabulary pools HKM targets; downstream numbers use one seed, so we treat sub-point differences as ties; imbalance is synthetic rather than naturally long-tailed (Liu et al., 2019); and our evaluators are supervised ResNet probes, a proxy for the SSL setting the method was built for. Whether the residual small-budget tail benefit compounds at larger scale, with multiple seeds and genuine SSL evaluation, is the next step.

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